

SPEECH TRANSCRIPTION

Using Raspberry Pi Zero

SOMNATH BERA

In a previous initiative, the Speech GPT terminal was introduced, which allowed users to ask questions in the terminal and receive spoken answers. This device further enhances the design using a more powerful board to transcribe audio in real time. Audio transcription is useful for recording meeting minutes, converting lectures to text, and various other applications. This device can also aid journalists in transcribing interviews. Here, GPT and a microphone are employed for transcription.

Raspberry Pi Zero is utilised for speech recognition, a form of artificial intelligence programming. Understanding and reproducing phonetics requires significant training and refinement. However, some free open source software, including Google-STT, can be used for this purpose.

The author's prototype is shown in Fig. 1. The components needed to build this device are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram for speech transcription on Raspberry Pi Zero. It is built using Raspberry Pi Zero, PAM8403 amplifier, two speakers, and a few other components.

There is no sound output on the Pi Zero board. An HDMI device (for example, a TV) must be connected



Fig. 1: Author's prototype

BILL OF MATERIALS

Components	Quantity
Raspberry Pi Zero	1
PAM8403	1
USB microphone	1
OTG cable for connecting USB microphone	1
Capacitor C1, C2 (0.1 μ F)	2
Capacitor C3, C4 (10 μ F, 16V)	2
Resistors R1, R2 (270-ohm)	2
Resistors (R3, R4 (150-ohm)	2
8-ohm, 0.5-watt speaker	2
On/off switch	1

to the Pi Zero to get sound. However, there is an easier way, as shown in the schematic. Any two of the four PWM GPIO pins 12, 13, 18, and 19 can be used to force audio output by adding just one line in the '/boot/config.txt' file - [dtoverlay=audre map.pins_18_13]. Open the file, go to the bottom, and add this line, followed by a reboot. Next time, the audio signal will be available on those

two GPIO pins. Just add a headphone to the IN-L and IN-R to get sound on the headphones. The PAM8403 amplifies the sound further for a speaker. Fig. 3 shows the image of the OTG cable for connecting the USB microphone.

```
$> sudo nano /boot/
config.txt
```

Add the above line and reboot. For Pi-5 the config.txt may be in another place. This

PAM8403 is rated for 5V but works on 3.3V too.

Software configuration

Two software packages will be experimented with here: pocketsphinx and sphcat. Installation of pocketsphinx is easy and straightforward, while sphcat is a command line program. Pocketsphinx is a Python pip module and can be used only within Python code.

1. pocketsphinx. This lightweight software works mainly on English phonetics.

```
$> pip install pocketsphinx
$>sudo apt-get install -y python3-
pocketsphinx pocketsphinx-en-us
```

Above is a small Python program for live speech transcription: from pocketsphinx import LiveSpeech

```
speech = LiveSpeech()
print("Listening...")
for phrase in speech:
    print(f"Transcript: {phrase}")
```

Run this code and then speak into the USB microphone attached